

3.7 How Science Works

How Science Works is an underpinning set of concepts and is the means whereby students come to understand how scientists investigate scientific phenomena in their attempts to explain the world about us. Moreover, *How Science Works* recognises the contribution scientists have made to their own disciplines and to the wider world.

It recognises that scientists may be influenced by their own beliefs and that these can affect the way in which they approach their work. Also, it acknowledges that scientists can and must contribute to debates about the uses to which their work is put and how their work influences decision-making in society.

In general terms, it can be used to promote students' skills in solving scientific problems by developing an understanding of:

- the concepts, principles and theories that form the subject content
- the procedures associated with the valid testing of ideas and, in particular, the collection, interpretation and validation of evidence
- the role of the scientific community in validating evidence and also in resolving conflicting evidence.

As students become proficient in these aspects of *How Science Works*, they can also engage with the place and contribution of science in the wider world. In particular, students will begin to recognise

- the contribution that scientists, as scientists, can make to decision-making and the formulation of policy
- the need for regulation of scientific enquiry and how this can be achieved
- how scientists can contribute legitimately to debates about those claims which are made in the name of science.

An understanding of *How Science Works* is a requirement for this specification and is set out in the following points which are taken directly from the *GCE AS and A Level subject criteria for science subjects*. Each point is expanded in the context of Biology. The specification references given illustrate where the example is relevant and could be incorporated.

A	Use theories, models and ideas to develop and modify scientific explanations
	Scientists use theories and models to attempt to explain observations. These theories and models can form the basis for scientific experimental work. Scientific progress is made when validated evidence is found that supports a new theory or model.
	<i>Examples in this specification include:</i> • Theoretical models such as the mass transport of water, 3.1.3 • Physical models such as the use of the lock and key model to explain the properties of enzymes, 3.1.2 • Mathematical models, such as the Hardy-Weinberg principle, predict that allele frequencies do not change from generation to generation 3.4.8
B	Use knowledge and understanding to pose scientific questions, define scientific problems, present scientific arguments and scientific ideas
	Scientists use their knowledge and understanding when observing objects and events, in defining a scientific problem and when questioning their own explanations or those of other scientists. Scientific progress is made when scientists contribute to the development of new ideas, materials and theories.
	<i>Examples in this specification include:</i> • pose scientific questions and define scientific problems in Investigative and practical skills